

HAT-P-12b

R-0741

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_250128 · created 2026-07-28T01:58:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460703.9296 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1791 +/- 0.0088
Transit depth (Rp/Rs)^2	3.21 +/- 0.31 [%]
Orbital Inclination (inc)	86.3 +/- 0.5
Airmass coefficient 1 (a1)	0.958 +/- 0.013
Airmass coefficient 2 (a2)	0.0283 +/- 0.0087
Scatter in the residuals of the lightcurve fit is	1.37 %
Best Comparison Star	#2 - [243, 225]
Optimal Method	PSF photometry
Transit Duration (day)	0.0775 +/- 0.008

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1791 ± 0.0088 vs 0.1406 (deviation 3.24σ)
Duration fit vs published	0.0775 d vs 0.0975 d (deviation 1.85σ)
Mid-time O–C	-11.2 min
Depth SNR	15.1
Residual scatter	1.37 %

Light curve

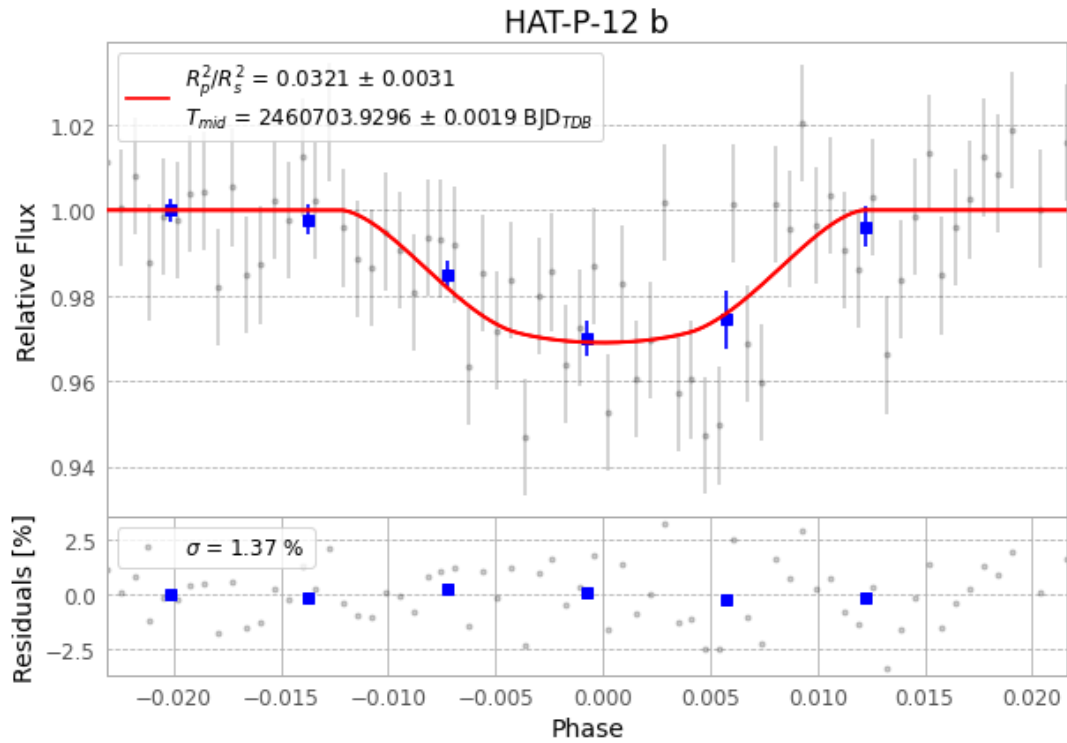


Figure 1 — FinalLightCurve_HAT-P-12 b_2025-01-28.png (SHA-256 4b5ccf58d8a02df2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 185], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 759b5c0143229d7a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), HATP-12250128082715.FITS → HATP-12250128115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-250128.log (46db4a1201c9...), FinalLightCurve_HAT-P-12 b_2025-01-28.png (4b5ccf58d8a0...), FinalLightCurve_HAT-P-12 b_2025-01-28.csv (0d273c629bd7...)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 01:58Z.